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MySQL ALTER TABLE Statement

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MySQL ALTER TABLE Statement

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The `ALTER TABLE` statement is used to add, delete, or modify columns in an existing table.

The `ALTER TABLE` statement is also used to add and drop various constraints on an existing table.

Common `ALTER TABLE` operations are:

- Add column - Adds a new column to a table
- Drop column - Deletes a column in a table
- Rename column - Renames a column
- Modify column - Changes the data type, size, or constraints of a column
- Add constraint - Adds a new constraint
- Rename table - Renames a table

Syntax

```
ALTER TABLE table_name
ADD column_name datatype;
```

The following SQL adds an "Email" column to the "Customers" table:

Example

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```
ALTER TABLE Customers
ADD Email varchar(255);
```

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Syntax

```
ALTER TABLE table_name
DROP COLUMN column_name;
```

The following SQL deletes the "Email" column from the "Customers" table:

Example

```
ALTER TABLE Customers
DROP COLUMN Email;
```

ALTER TABLE - RENAME COLUMN

To rename a column in a table, use the following syntax:

Syntax

```
ALTER TABLE table_name
RENAME COLUMN old_name to new_name;
```

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ALTER TABLE - MODIFY Datatype

To modify the data type, size or constraints of a column in a table, use the following syntax:

Syntax

```
ALTER TABLE table_name
MODIFY column_name new_datatype constraint;
```

The following SQL modifies the size of the "Email" column to varchar(100), and we also add a `NOT NULL` constraint:

Example

```
ALTER TABLE Customers
MODIFY Email varchar(100) NOT NULL;
```

ALTER TABLE - ADD CONSTRAINT

To add a constraint to an existing table, use the following syntax:

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```
ALTER TABLE table_name
ADD CONSTRAINT constraint_name constraint_definition;
```

The following SQL adds a constraint named "CHK_Age" that is a `CHECK` constraint that ensures that the "Age" column has a value of 18 and above:

Example

```
ALTER TABLE Members
ADD CONSTRAINT CHK_Age CHECK (Age >= 18);
```

ALTER TABLE - Rename table

To rename a table, use the following syntax:

Syntax

```
ALTER TABLE table_name
RENAME TO new_table_name;
```

The following SQL renames the "Customers" table to "Clients":

Example

```
ALTER TABLE Customers
RENAME TO Clients;
```

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Assume we have a "Persons" table, that looks like this:

ID	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

Now we want to add a column named "DateOfBirth" in the "Persons" table.

We use the following SQL statement:

Example

```
ALTER TABLE Persons
ADD DateOfBirth date;
```

Notice that the new column, "DateOfBirth", is of type date and is going to hold a date. The data type specifies what type of data the column can hold. For a complete reference of all the data types available in MySQL, go to our Data Types reference.

The "Persons" table will now look like this:

ID	LastName	FirstName	Address	City	DateOfBirth
1	Hansen	Ola	Timoteivn 10	Sandnes	
2	Svendson	Tove	Borgvn 23	Sandnes	
3	Pettersen	Kari	Storgt 20	Stavanger	

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Now we want to change the data type of the column named "DateOfBirth" in the "Persons" table.

We use the following SQL statement:

Example

```
ALTER TABLE Persons
MODIFY COLUMN DateOfBirth year;
```

Notice that the "DateOfBirth" column is now of type year and is going to hold a year in a two- or four-digit format.

DROP COLUMN Example

Next, we want to delete the column named "DateOfBirth" in the "Persons" table.

We use the following SQL statement:

Example

```
ALTER TABLE Persons
DROP COLUMN DateOfBirth;
```

The "Persons" table will now look like this:

ID	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes

‹ Previous

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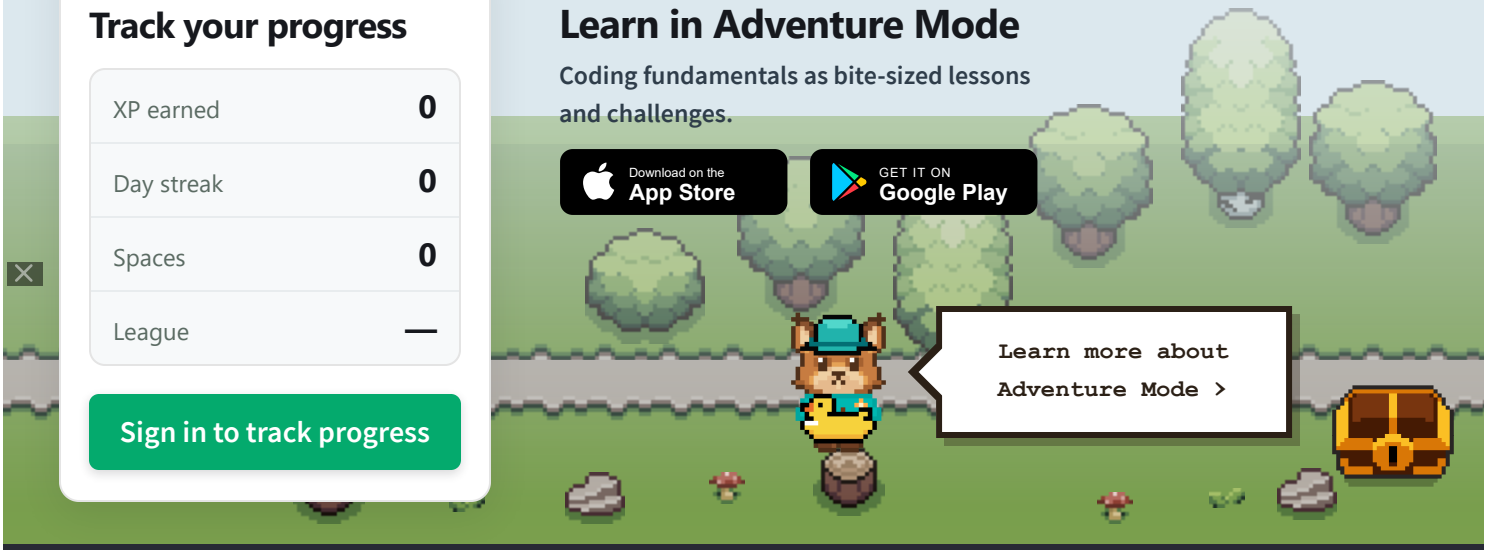
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